

Lab04 - Odometry

MTRN3100 - UNSW School of Mechanical and Manufacturing Engineering

Task 1 - System parameters

There are two unknown parameters in the kinematic model of the differential robot. Those are the wheel radius and the axle length.

Questions

- **Question 1:** Find the radius of the wheels.
- **Question 2:** Find the axle length of the demo bot.

Task 2 - Encoder Odometry

Open and complete the EncoderOdometry.hpp file. Verify that your odometry code works correctly by pushing the robot and comparing how far the robot has gone and the calculated distance. You do not need to drive the robot for this task, and can simply push it. Ensure that forwards, backward, and turning work.

Questions

- **Question 3:** What is odometry?
- **Question 4:** Why do we calculate and use odometry?
- **Question 5:** What could make wheel odometry inaccurate?

Task 3 - IMU Odometry

Open and complete the IMUOdometry.hpp file. You should notice that the IMU appears to work for a short period before it drifts significantly. You do not need to drive the robot for this task, and can simply push it.

Note: Do not spend too long implementing IMU odometry. This task is meant to show why IMU odometry by itself is not a great solution.

- **Question 6:** Why does IMU odometry accumulate position error over time?
- **Question 7:** Although the IMU has poor performance for odometry, how can it still be useful? Consider what happens when a differential robot turns.

Task 4 - Setpoints

Using either a PID or Bang-Bang controller, create a program that drives the robot forward two wheel rotations, stops, then backward two wheel rotations. Use either of the wheel encoders to count the rotations. Use a moving average filter to detect when the system has come to rest before setting the next setpoint and activating the controller.

Your approach should start and end at the same location.

- **Question 8:** Why is it important to use a moving average filter to detect when the setpoint has been reached, rather than stopping as soon as the setpoint is first reached?